

Quantum Machine Learning with Neural Networks

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Outline

Quantum Machine Learning Landscape

- Quantum Data and Classical Data: $|\psi\rangle$ vs vectors $\mathbf{x} \in \mathbb{R}^n$
- Categories:
 - Quantum-enhanced classical ML
 - Quantum-native ML algorithms
 - Hybrid quantum-classical ML
- Resource-aware quantum learning models

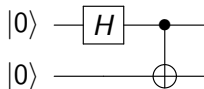
Quantum States and Gates

- Qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$
- Common Gates:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Entanglement via CNOT and multi-qubit systems

Quantum Circuit Example



- This circuit creates a Bell state: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- Essential for encoding correlations

Classical Neural Networks Review

- Universal approximation theorem
- Layer-wise structure: $y = \sigma(Wx + b)$
- Deep networks and expressivity
- Limitations in high-dimensional feature space

What is a QNN?

- Quantum analog of classical NNs using parameterized quantum circuits (PQC)
- Data encoding layer + entangling layer + variational layer
- Output is measurement of observables
- Trainable parameters: gate angles θ

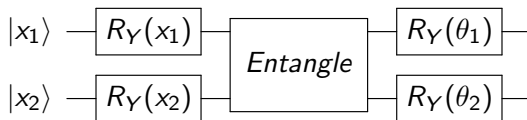
Generic QNN Architecture

qnn_diagram.png

Variational Quantum Circuits

- PQCs parameterized by θ
- Optimization via classical optimizer (e.g., COBYLA, SPSA, Adam)
- Objective: minimize loss $L(\theta)$ based on expectation values
- Cost function: $C(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$

Circuit Example: Ansatz Layer



- Hybrid encoding: data x_i and learnable parameters θ_i
- Can be scaled for deeper architectures

- Gradient-based: Parameter Shift Rule

$$\frac{\partial \langle O \rangle}{\partial \theta} = \frac{\langle O \rangle_{\theta+\pi/2} - \langle O \rangle_{\theta-\pi/2}}{2}$$

- Cost landscapes can suffer from barren plateaus
- Mitigation via tailored ansatz or local cost functions

Applications of QML + NN

- Quantum-enhanced image and speech classification
- Anomaly detection in quantum systems
- Quantum generative adversarial networks (QGANs)
- Quantum autoencoders for data compression

Challenges in QNNs

- Hardware constraints: decoherence, limited qubits
- Barren plateau problem in deep PQCs
- Lack of general-purpose quantum optimizers
- Scaling to larger problem instances

Future Directions

- Efficient hybrid architectures
- Adaptive circuit learning
- Near-term QML benchmarks
- Fault-tolerant QML pipelines